

Set Operations: Solved Questions + Practice Problems

Union, Intersection, Difference, Complements, De Morgan's Laws

Example Questions

Solved Example

1) Finite sets: compute $A \cap B$.

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 6\}$. Find $A \cap B$.

Solution. Intersection means “elements common to both”. Common elements are 3 and 4, hence

$$A \cap B = \{3, 4\}.$$

Solved Example

2) Finite sets: compute $A \cup B$.

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3\}$, $B = \{3, 4, 6\}$. Find $A \cup B$.

Solution. Union means “in A or in B (or both)”. Collect without repetition:

$$A \cup B = \{1, 2, 3, 4, 6\}.$$

Solved Example

3) Finite sets: compute $A \setminus B$.

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 6\}$. Find $A \setminus B$.

Solution. $A \setminus B$ means “in A but not in B ”. Remove 3 from A (since $3 \in B$):

$$A \setminus B = \{1, 2, 4\}.$$

Solved Example

4) Finite sets: compute $B \setminus A$.

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 4\}$, $B = \{2, 3, 5, 6\}$. Find $B \setminus A$.

Solution. Keep elements in B that are not in A . Since $2 \in A$, remove it:

$$B \setminus A = \{3, 5, 6\}.$$

Solved Example

5) Complements in a finite universe: compute A^c .

Let $U = \{a, b, c, d, e, f\}$ and $A = \{a, c, f\}$. Find A^c .

Solution. $A^c = U \setminus A$:

$$A^c = \{b, d, e\}.$$

Solved Example

6) Complements in a finite universe: compute B^c .

Let $U = \{1, 2, 3, 4, 5, 6, 7\}$ and $B = \{2, 3, 7\}$. Find B^c .

Solution. Remove the elements of B from U :

$$B^c = \{1, 4, 5, 6\}.$$

Solved Example

7) Translate difference into $\cap$ and complement.

Show directly (by element reasoning) that $A \setminus B = A \cap B^c$.

Solution. For any $x \in U$,

$$x \in A \setminus B \iff (x \in A) \text{ and } (x \notin B).$$

But $x \notin B \iff x \in B^c$, so

$$x \in A \setminus B \iff (x \in A) \text{ and } (x \in B^c) \iff x \in A \cap B^c.$$

Hence $A \setminus B = A \cap B^c$.

Solved Example

8) Interval sets: compute $A \cap B$.

Let $U = \mathbb{R}$, $A = (-2, 5]$, $B = [1, 7)$. Find $A \cap B$.

Solution. The overlap starts at 1 (included) and ends at 5 (included):

$$A \cap B = [1, 5].$$

Solved Example

9) Interval sets: compute $A \cup B$.

Let $U = \mathbb{R}$, $A = (-2, 5]$, $B = [1, 7)$. Find $A \cup B$.

Solution. Together they cover from -2 (not included) up to 7 (not included):

$$A \cup B = (-2, 7).$$

Solved Example

10) Interval sets: compute $A \setminus B$.

Let $U = \mathbb{R}$, $A = (-2, 5]$, $B = [1, 7)$. Find $A \setminus B$.

Solution. Remove the overlap $[1, 5]$ from A ; what remains is the left part:

$$A \setminus B = (-2, 1).$$

Solved Example

11) Complement in $\mathbb{R}$: compute A^c .

Let $U = \mathbb{R}$ and $A = [-1, 3)$. Find A^c .

Solution. Everything in $\mathbb{R}$ outside $[-1, 3)$:

$$A^c = (-\infty, -1) \cup [3, \infty).$$

Solved Example

12) Disjointness check.

Let $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$ with universe $U = \{1, 2, 3, 4, 5, 6\}$. Are A and B disjoint?

Solution. Compute the intersection:

$$A \cap B = \emptyset$$

since no number is both odd and even. Therefore A and B are disjoint.

Solved Example

13) A union with a subset situation.

Let $U = \{1, 2, 3, 4, 5\}$, $A = \{2, 3\}$, $B = \{1, 2, 3, 4\}$. Compute $A \cup B$.

Solution. Since every element of A is already in B , we have $A \cup B = B$:

$$A \cup B = \{1, 2, 3, 4\}.$$

Solved Example

14) A intersection with a subset situation.

Let $U = \{1, 2, 3, 4, 5\}$, $A = \{2, 3\}$, $B = \{1, 2, 3, 4\}$. Compute $A \cap B$.

Solution. If $A \subseteq B$, then $A \cap B = A$. Indeed,

$$A \cap B = \{2, 3\}.$$

Solved Example

15) Mixed expression: compute $(A \cup B) \setminus C$.

Let $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $C = \{2, 5, 7\}$. Find

$$(A \cup B) \setminus C.$$

Solution. First union:

$$A \cup B = \{1, 2, 3, 4, 5\}.$$

Now remove elements of C (remove 2 and 5; 7 is not present anyway):

$$(A \cup B) \setminus C = \{1, 3, 4\}.$$

Solved Example

16) Mixed expression: compute $A \cap (B \setminus C)$.

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 5, 6\}$, $C = \{3, 6\}$. Find $A \cap (B \setminus C)$.

Solution. First difference:

$$B \setminus C = \{2, 5\} \quad (\text{remove } 3, 6).$$

Now intersect with A :

$$A \cap (B \setminus C) = \{2\}.$$

Solved Example

17) De Morgan (first law) by “region logic” + diagram.

Explain why $(A \cup B)^c = A^c \cap B^c$.

Solution. The set $(A \cup B)^c$ consists of points in U that are *not* in $A \cup B$. That means a point is not in A *and* not in B at the same time:

$$x \in (A \cup B)^c \iff x \notin A \cup B \iff (x \notin A) \text{ and } (x \notin B) \iff x \in A^c \cap B^c.$$

So the regions match exactly: “outside both circles” is the same as “in A^c and in B^c ”.

Solved Example

18) De Morgan (second law) full proof (what you asked).

Prove $(A \cap B)^c = A^c \cup B^c$ using $\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q)$.

Solution. Let $x \in U$. Then

$$x \in (A \cap B)^c \iff x \notin A \cap B \iff \neg(x \in A \cap B).$$

But $x \in A \cap B \iff (x \in A) \wedge (x \in B)$. Hence

$$\neg(x \in A \cap B) \iff \neg((x \in A) \wedge (x \in B)) \iff \neg(x \in A) \vee \neg(x \in B),$$

using $\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q)$. Now rewrite $\neg(x \in A)$ as $x \in A^c$ and $\neg(x \in B)$

as $x \in B^c$:

$$\neg(x \in A) \vee \neg(x \in B) \iff (x \in A^c) \vee (x \in B^c) \iff x \in A^c \cup B^c.$$

Therefore $(A \cap B)^c = A^c \cup B^c$.

Solved Example

19) Compute using De Morgan: find $(A \cup B)^c$ two ways.

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{4, 5, 8\}$.

Solution. Direct way:

$$A \cup B = \{1, 2, 3, 4, 5, 8\} \Rightarrow (A \cup B)^c = \{6, 7\}.$$

De Morgan way:

$$A^c = \{5, 6, 7, 8\}, \quad B^c = \{1, 2, 3, 6, 7\}.$$

Then

$$A^c \cap B^c = \{6, 7\},$$

which matches the direct method.

Solved Example

20) Compute using De Morgan: find $(A \cap B)^c$ two ways.

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4\}$.

Solution. Direct way:

$$A \cap B = \{2, 4\} \Rightarrow (A \cap B)^c = \{1, 3, 5, 6, 7, 8\}.$$

De Morgan way:

$$A^c = \{1, 3, 5, 7\}, \quad B^c = \{5, 6, 7, 8\},$$

so

$$A^c \cup B^c = \{1, 3, 5, 6, 7, 8\},$$

same answer.

20 Practice Problems (similar style)

Practice Problems

- P1.** Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 5\}$, $B = \{2, 4, 5, 6\}$. Find $A \cap B$.
- P2.** Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 3, 5\}$, $B = \{1, 2, 6\}$. Find $A \cup B$.
- P3.** Let $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 3, 4, 7\}$, $B = \{1, 3, 5\}$. Find $A \setminus B$.
- P4.** Let $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 3, 4, 7\}$, $B = \{1, 3, 5\}$. Find $B \setminus A$.
- P5.** Let $U = \{a, b, c, d, e, f, g\}$, $A = \{a, d, g\}$. Find A^c .
- P6.** Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $B = \{2, 3, 6, 8\}$. Find B^c .
- P7.** Prove (by element-chasing): $A \setminus B = A \cap B^c$.
- P8.** In $U = \mathbb{R}$, let $A = (-5, 2]$ and $B = [-1, 6)$. Find $A \cap B$.
- P9.** In $U = \mathbb{R}$, let $A = (-5, 2]$ and $B = [-1, 6)$. Find $A \cup B$.
- P10.** In $U = \mathbb{R}$, let $A = (-5, 2]$ and $B = [-1, 6)$. Find $A \setminus B$.
- P11.** In $U = \mathbb{R}$, let $A = [0, 4)$; find A^c .
- P12.** Let $A = \{x \in \mathbb{Z} : x \text{ is even}\}$ and $B = \{x \in \mathbb{Z} : x \text{ is odd}\}$. Are A and B disjoint? Justify.
- P13.** Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 3\}$, $B = \{1, 2, 3, 4, 5\}$. Compute $A \cup B$ and $A \cap B$.
- P14.** Let $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $C = \{1, 5, 7\}$. Find $(A \cup B) \setminus C$.
- P15.** Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3, 6\}$, $B = \{2, 4, 6\}$, $C = \{6\}$. Find $A \cap (B \setminus C)$.
- P16.** Prove De Morgan (first law): $(A \cup B)^c = A^c \cap B^c$.
- P17.** Prove De Morgan (second law): $(A \cap B)^c = A^c \cup B^c$.
- P18.** Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3\}$, $B = \{3, 4, 8\}$. Compute $(A \cup B)^c$ two ways.
- P19.** Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{2, 4, 6, 8\}$, $B = \{2, 3, 4, 5\}$. Compute $(A \cap B)^c$ two ways.
- P20.** (Challenge) Prove: $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.